Fundamentals of Metal Forming

Groover M.P.: Chapter 14-15

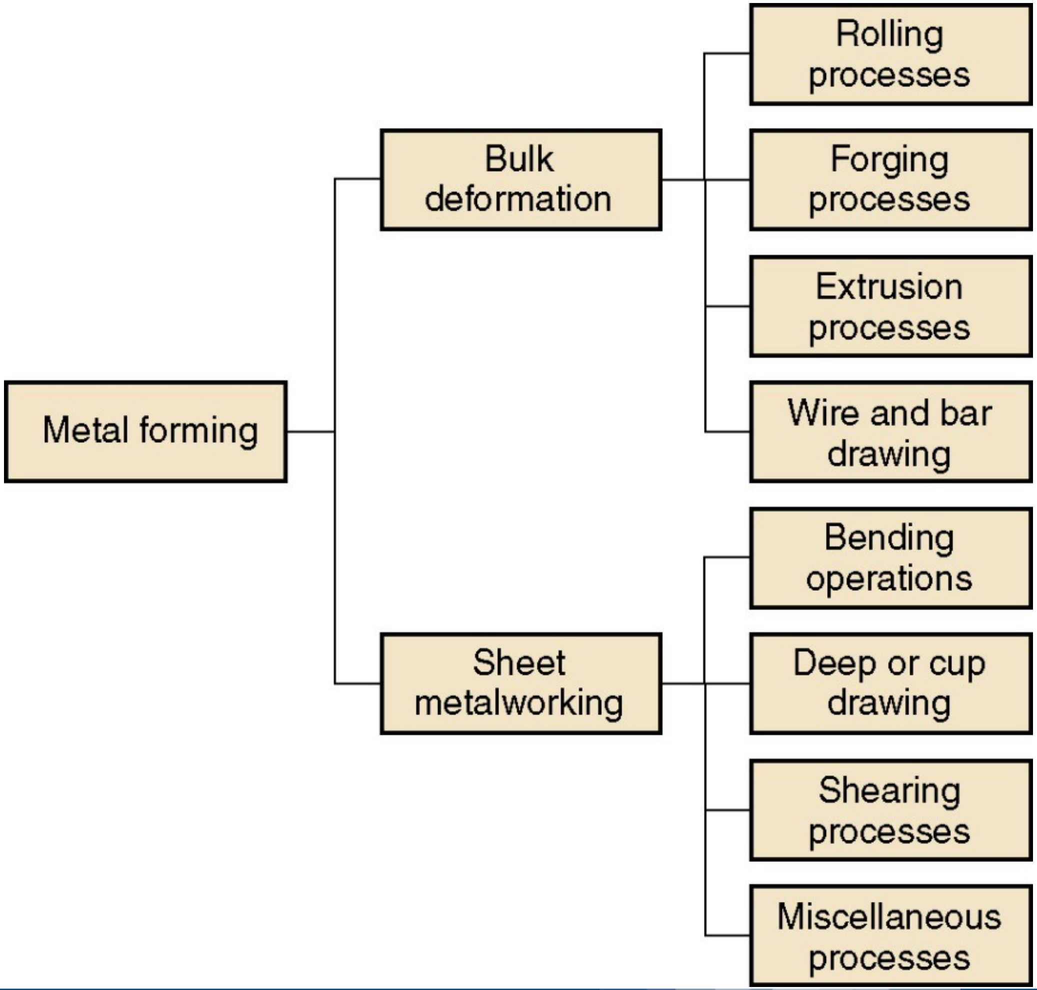
Learning Targets

The student:

- has an overview of metal forming processes;
- knows about the fundamental principles, rules, and laws that can be used to describe the deformation of solid materials;
- applies these principles, rules, and laws in a simple but relevant case: open-die forging.

METAL FORMING PROCESSES

Large group of manufacturing processes in which plastic deformation is used to change the shape of metal workpiece.



BULK DEFORMATION PROCESSES

Bulk deformation processes are characterized by significant deformations and massive shape changes.

"**Bulk**" refers to workpieces with relatively **low surface area-to-volume ratios**.

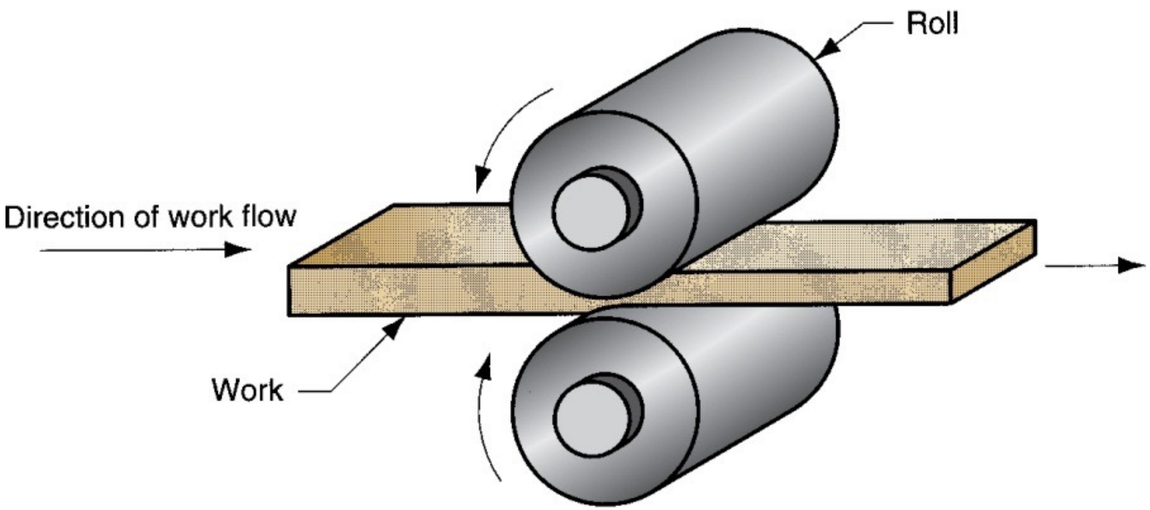
Starting work shapes are usually simple geometries, e.g. cylindrical billets or rectangular bars.

Bulk deformation processes:

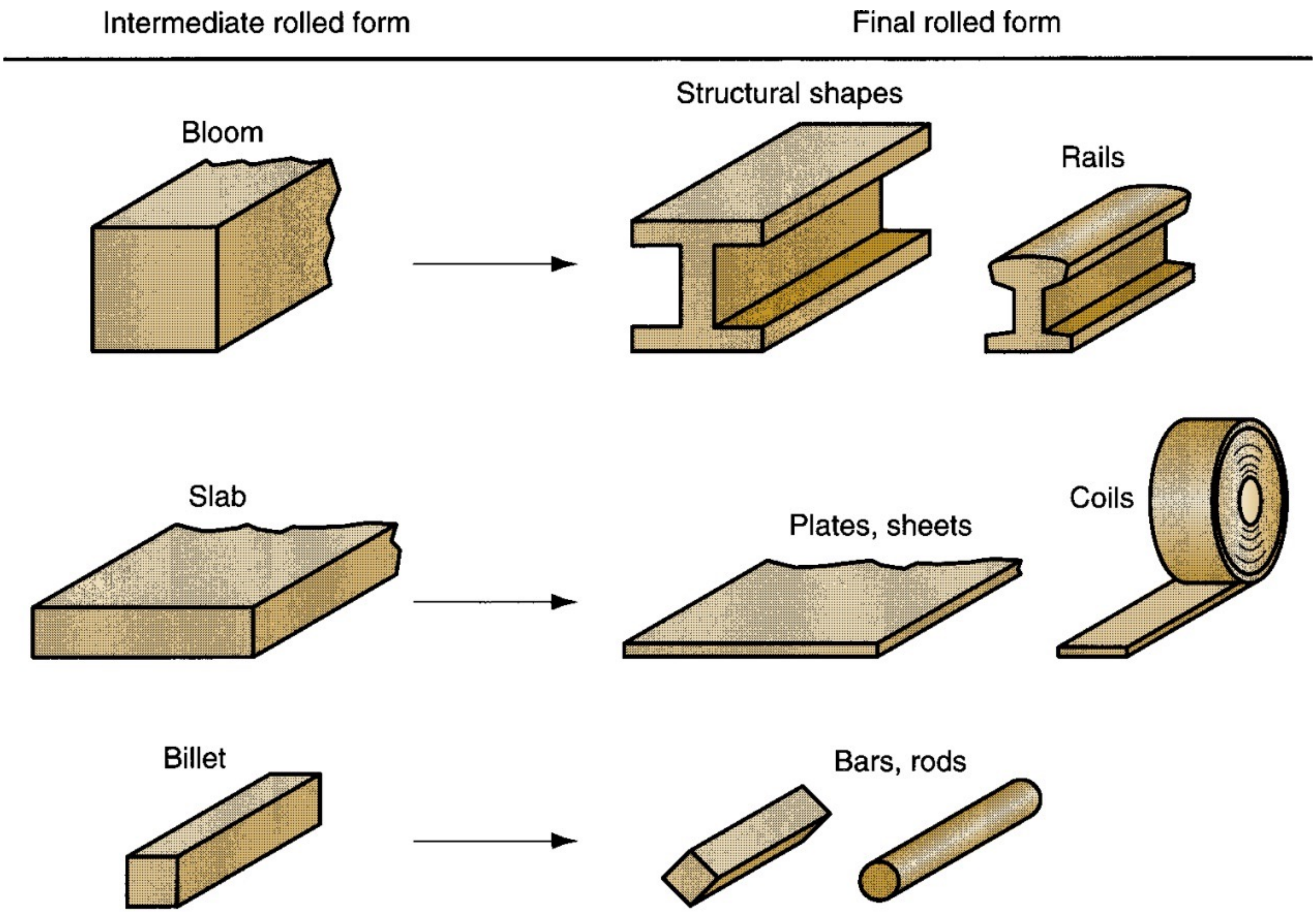
- Rolling
- Forging
- Extrusion
- Wire and bar drawing

ROLLING

Deformation process in which work thickness is reduced by compressive forces exerted by two opposing rolls (shown below is flat rolling).



ROLLED PRODUCTS MADE OF STEEL



ROLLING

https://youtu.be/6xnKmt_gsLs

FORGING

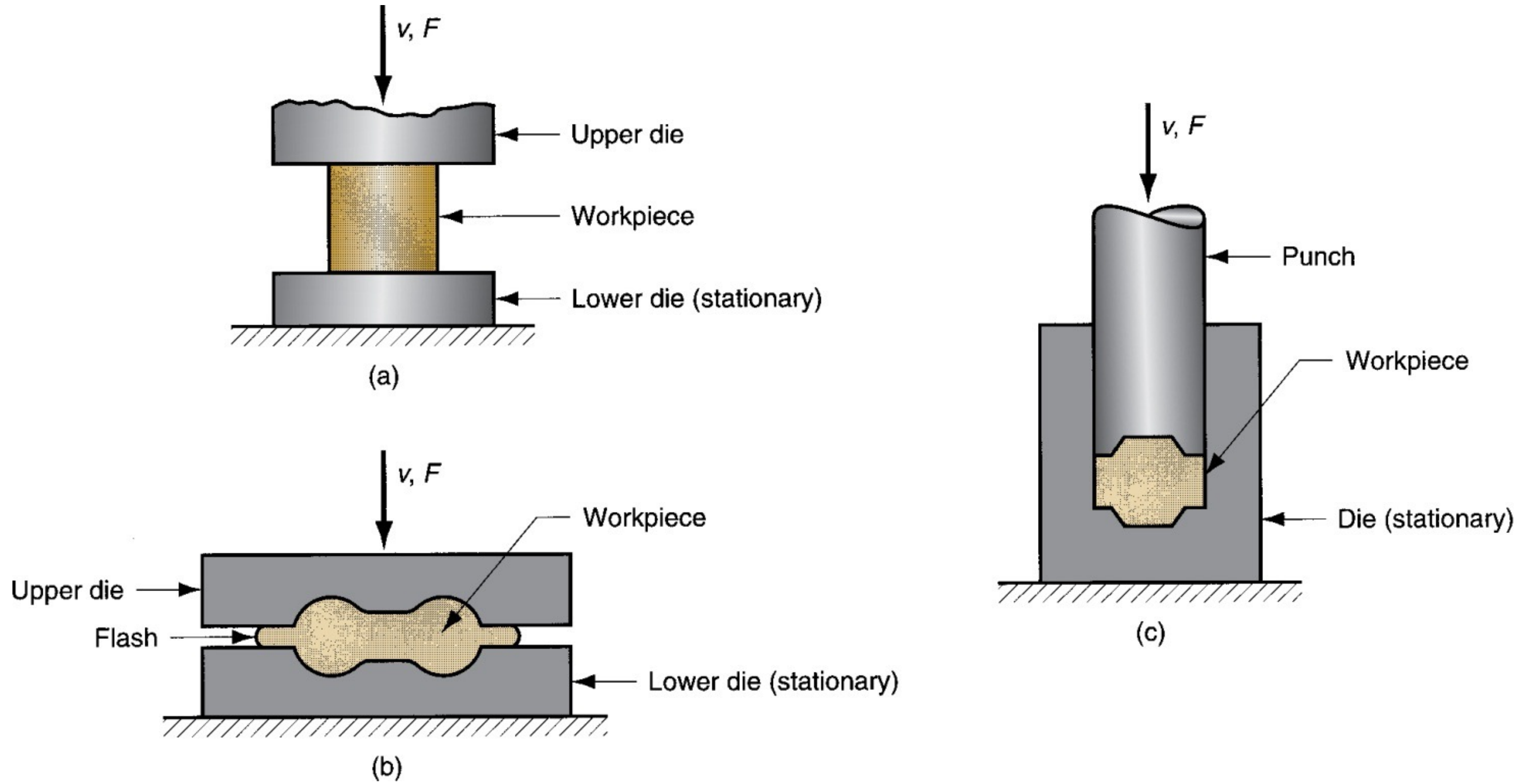
Deformation process in which work is compressed between two dies.

- Oldest of the metal forming operations: dates from about 5000 B.C.
- Products: engine crankshafts, connecting rods, gears, jet engine turbine parts.

Types of forging operations:

- ***Open-die forging***: work is compressed between two flat dies, allowing metal to flow laterally with minimum constraint.
- ***Impression-die forging***: die contains cavity or impression that is imparted to workpiece. Metal flow is constrained so that flash is created.
- Flashless forging: workpiece is completely constrained in die. No excess flash is created.

TYPES OF FORGING OPERATIONS



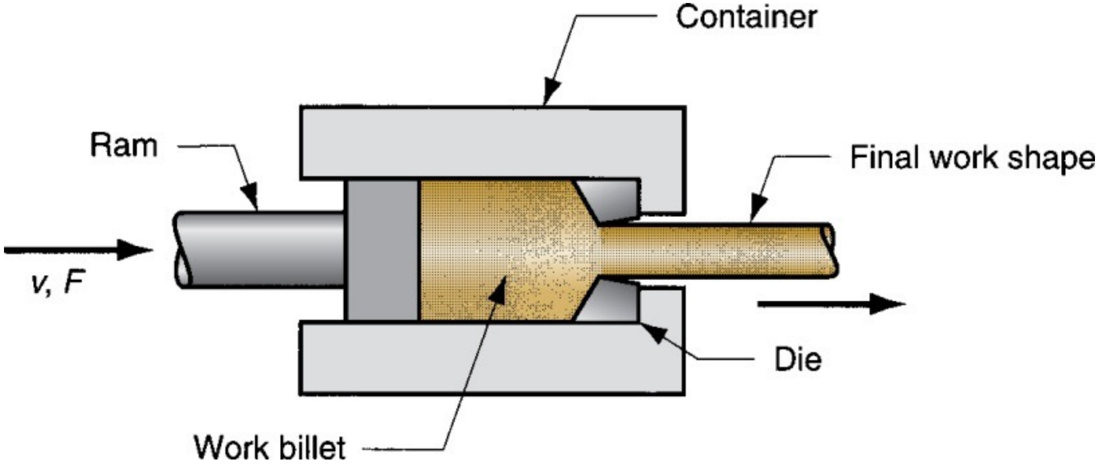
(a) Open-die forging, (b) Impression-die forging, and (c) Flashless forging.

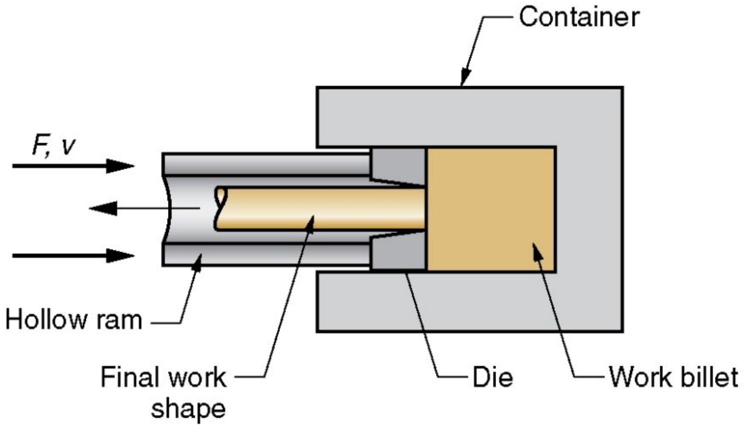
FORGING

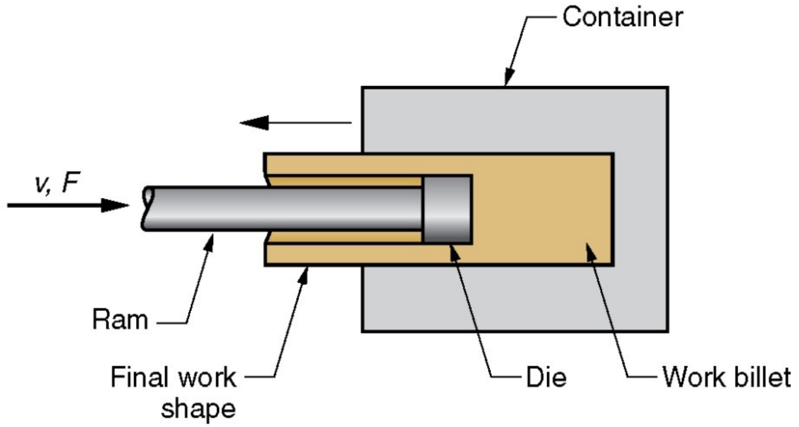
EXTRUSION

Compression forming process in which work **metal is forced to flow through a die opening to produce a desired cross-sectional shape**. In general, extrusion is used to produce long parts of uniform cross sections.

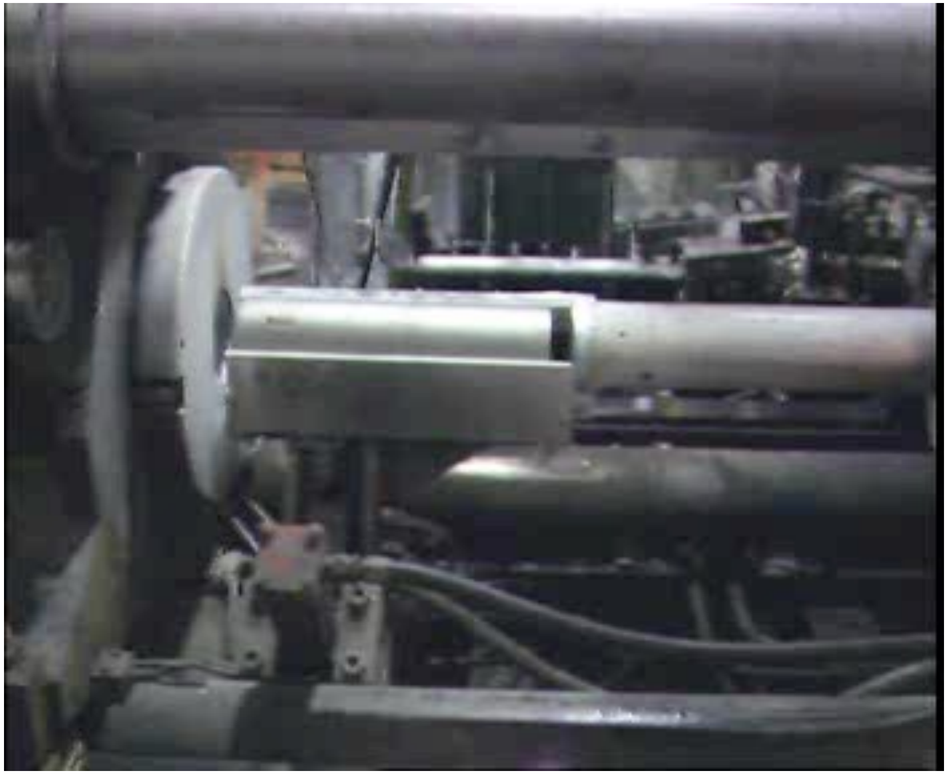
Two basic types: Direct extrusion, Indirect extrusion.







EXTRUSION



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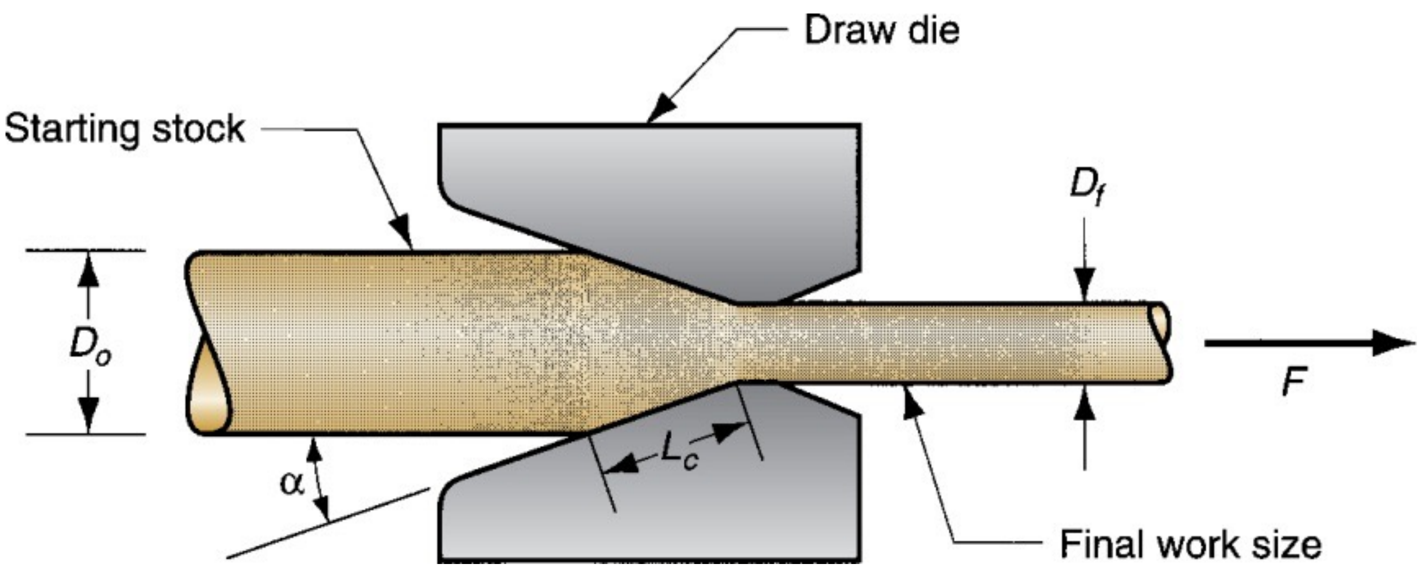
Wire and Bar Drawing

Cross section of a bar, rod, or wire is reduced by pulling it through a die opening.

- Similar to extrusion except work is *pulled* through die in drawing, it is *pushed* through in extrusion.
- Although drawing applies tensile stress, compression also plays a significant role since metal is squeezed as it passes through die opening.



Sheet Metalworking

Forming and related operations performed on metal sheets, strips, and coils.

High surface area-to-volume ratio of starting metal, which distinguishes these from bulk deformation.

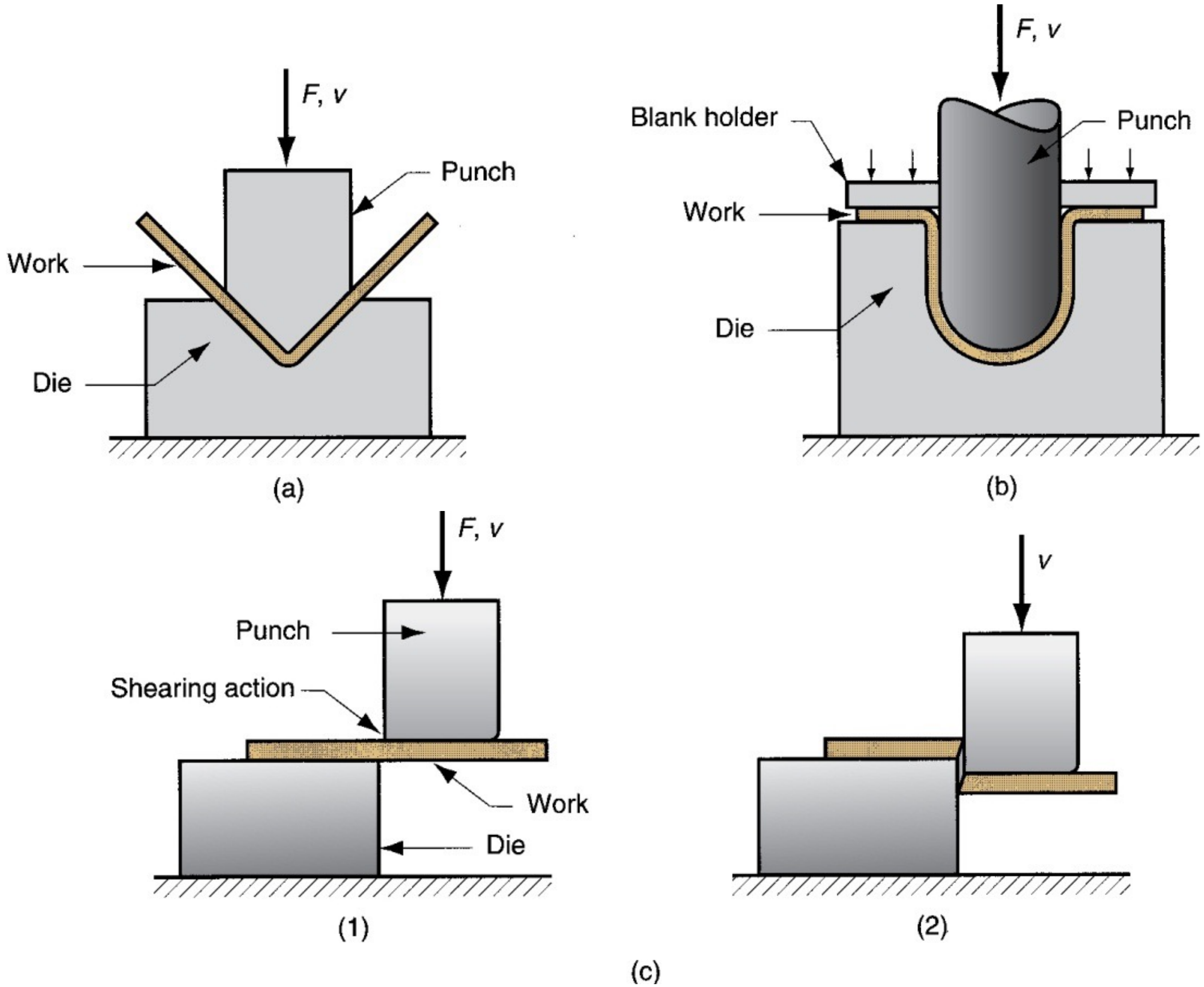
Often called ***press-working*** because these operations are performed on presses:

- Parts are called *stampings*
- Usual tooling: *punch* and *die*

Sheet metalworking processes:

- Bending operations
- Deep drawing
- Shearing processes

SHEET METALWORKING



(a) Bending and (b) Deep drawing

(c) Shearing: (1) punch first contacting sheet and (2) after cutting

Sheet Metalworking

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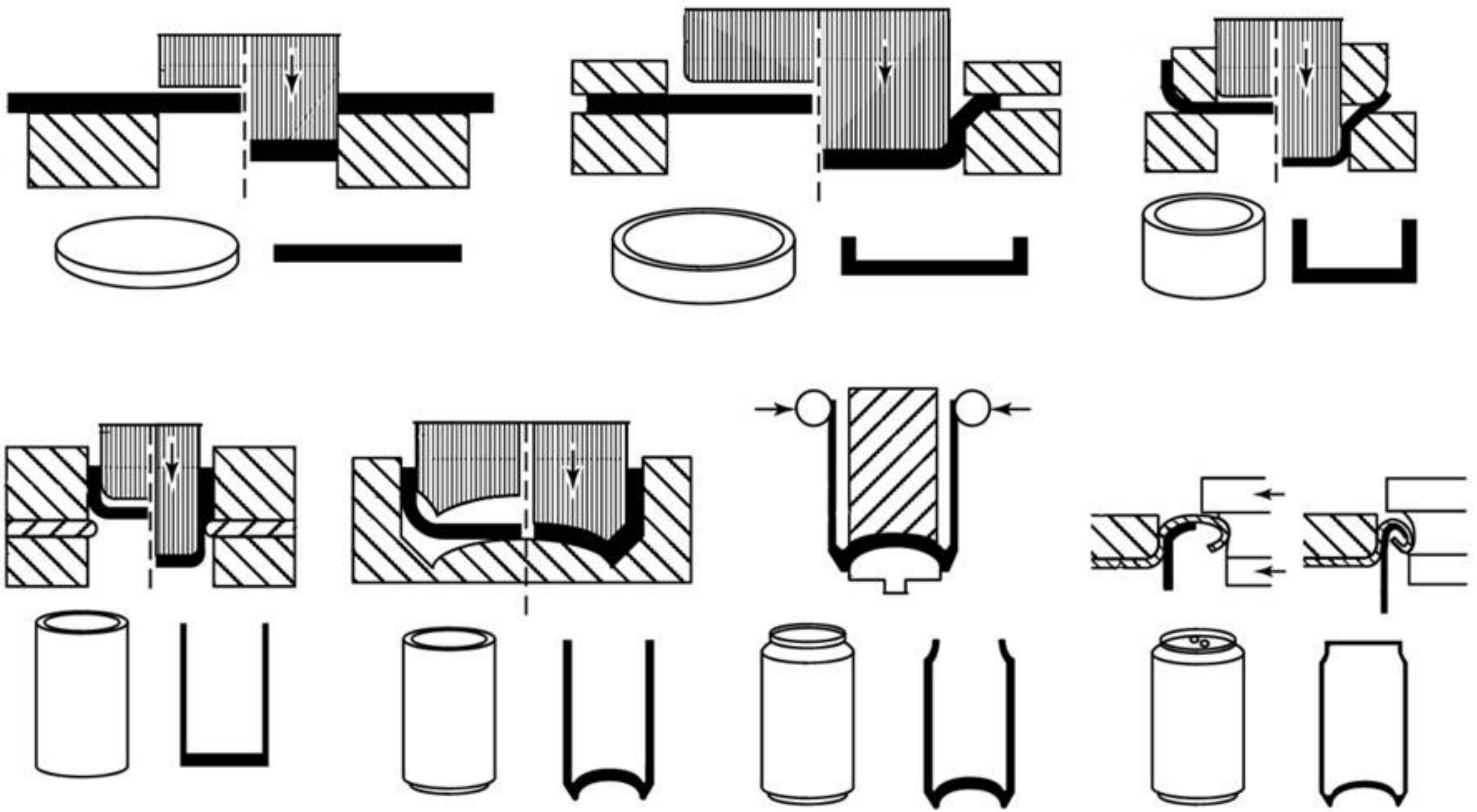
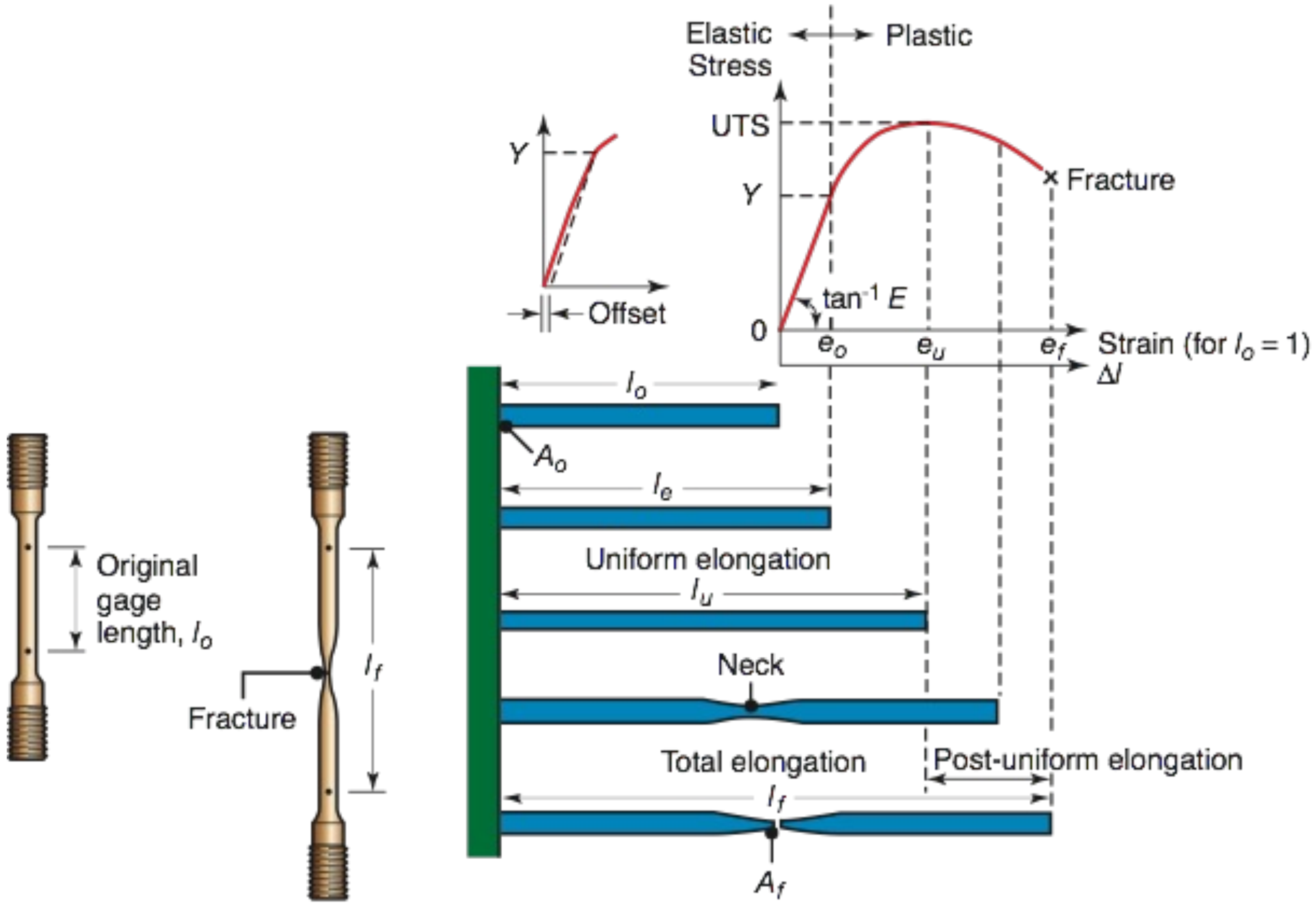
SHEET METALWORKING



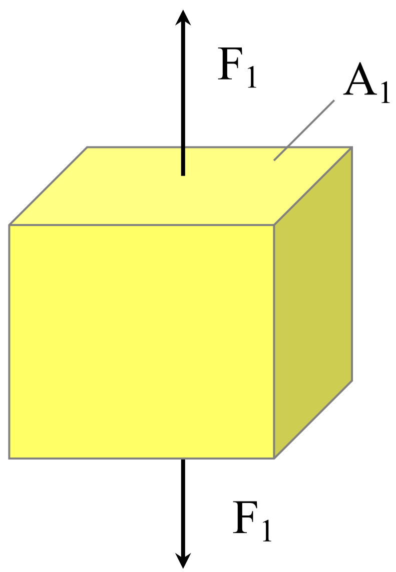
SHEET METALWORKING

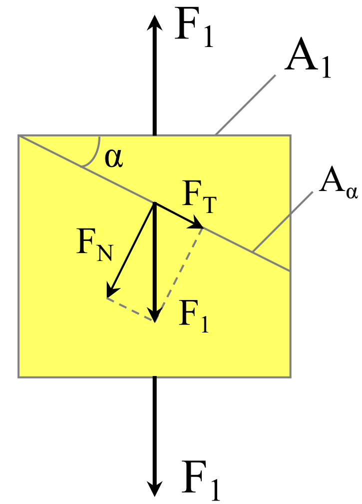
Tensile Test



TENSILE TEST

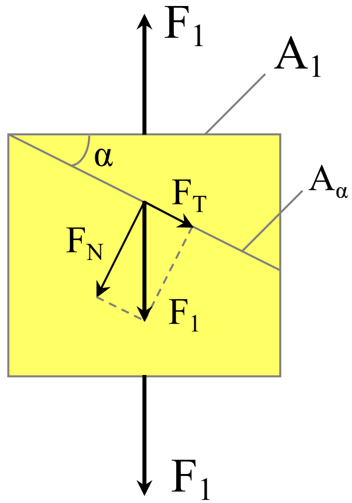
In tensile test we considered a uniaxial stress.
What happens if we change the cross-sectional orientation with respect to the original one (orthogonal to the load direction)?





TENSILE TEST

Let's consider a section rotated by α with respect to the original one (orthogonal to the load direction):



Forces:

$$F_N = F_1 \cos\alpha \quad ; \quad F_T = F_1 \sin\alpha$$

Normal stress:

$$\sigma_\alpha = \frac{F_N}{A_\alpha} = \frac{F_1 \cos\alpha}{A_1 / \cos\alpha} = \sigma \cos^2\alpha$$

Shear stress:

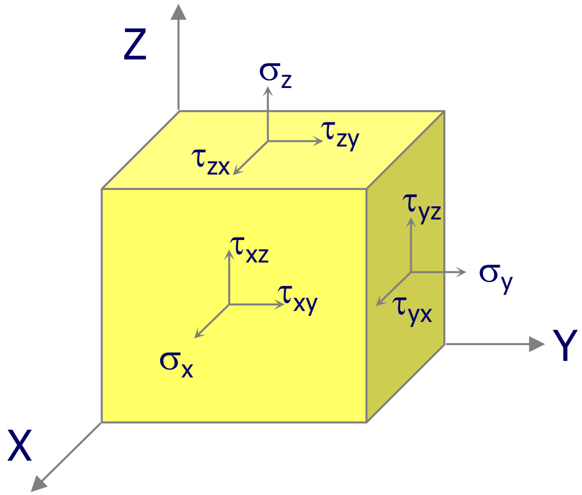
$$\tau_\alpha = \frac{F_T}{A_\alpha} = \frac{F_1 \sin\alpha}{A_1 / \cos\alpha} = \sigma \sin\alpha \cos\alpha$$

$$\tau_\alpha = \sigma \frac{1}{2} \sin 2\alpha$$

$$\tau_{max} = \tau_{\alpha=45^\circ} = \sigma/2$$

Deformation of Solid Materials

Let's consider a infinitesimal cube of material. The most general state of stress at a point may be represented by the following 3D stress matrix:



$$\begin{bmatrix} \sigma_x & \tau_{yx} & \tau_{zx} \\ \tau_{xy} & \sigma_y & \tau_{zy} \\ \tau_{xz} & \tau_{yz} & \sigma_z \end{bmatrix}$$

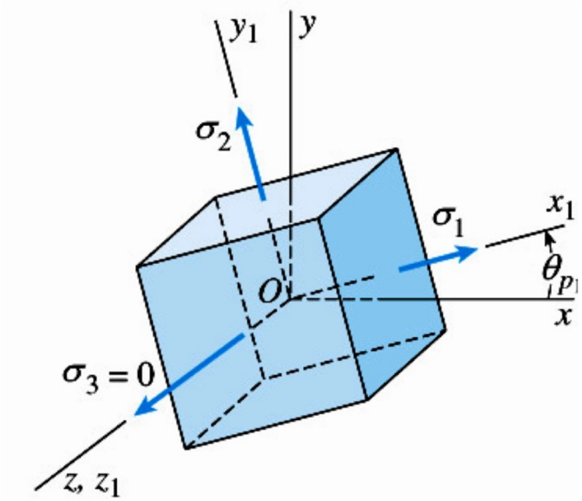
From moment equilibrium conditions on the infinitesimal stress cube, it may be shown that in magnitude:

$$\begin{cases} \tau_{xy} = \tau_{yx} \\ \tau_{yz} = \tau_{zy} \\ \tau_{zx} = \tau_{xz} \end{cases}$$

This is known as the shear stress reciprocity.

DEFORMATION OF SOLID MATERIALS

Same state of stress is represented by a different set of components if axes are rotated.
There is a special set of components (when axes are rotated) where all the shear components are zero. In this condition the axes are called **principal axes** and the normal stresses are called **principal stresses**.



Deformation of Solid Materials

Let's consider a infinitesimal cube of material.
The most general state of stress-strain at a point may be represented by the following 3D stress matrix and 3D strain matrix:

$$\begin{bmatrix} \sigma_x & \tau_{yx} & \tau_{zx} \\ \tau_{xy} & \sigma_y & \tau_{zy} \\ \tau_{xz} & \tau_{yz} & \sigma_z \end{bmatrix} \text{ and } \begin{bmatrix} \varepsilon_x & \varepsilon_{yx} & \varepsilon_{zx} \\ \varepsilon_{xy} & \varepsilon_y & \varepsilon_{zy} \\ \varepsilon_{xz} & \varepsilon_{yz} & \varepsilon_z \end{bmatrix}$$

If we consider the principal axes, the most general state of stress-strain at a point may be represented by the following diagonal matrices:

$$\begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix} \text{ and } \begin{bmatrix} \varepsilon_1 & 0 & 0 \\ 0 & \varepsilon_2 & 0 \\ 0 & 0 & \varepsilon_3 \end{bmatrix}$$

Deformation of Solid Materials

Stress-strain states:

- Triaxial stress: $\sigma_1 \neq 0; \quad \sigma_2 \neq 0; \quad \sigma_3 \neq 0;$
- Plane stress: $\sigma_1 \neq 0; \quad \sigma_2 = 0; \quad \sigma_3 \neq 0;$
- Uniaxial stress: $\sigma_1 \neq 0; \quad \sigma_2 = 0; \quad \sigma_3 = 0;$
- Triaxial strain: $\varepsilon_1 \neq 0; \quad \varepsilon_2 \neq 0; \quad \varepsilon_3 \neq 0;$
- Plane strain: $\varepsilon_1 > 0; \quad \varepsilon_2 = 0; \quad \varepsilon_3 < 0;$

Elastic Region

Hooke's Law for triaxial stress

$$\varepsilon_1 = \frac{1}{E}\left[\sigma_1 - \nu\left(\sigma_2 + \sigma_3\right)\right]$$

$$\varepsilon_2 = \frac{1}{E}\left[\sigma_2 - \nu\left(\sigma_3 + \sigma_1\right)\right]$$

$$\varepsilon_3 = \frac{1}{E}\left[\sigma_3 - \nu\left(\sigma_1 + \sigma_2\right)\right]$$

Poisson's ratio: $\nu = -\frac{lateral\ strain}{axial\ strain}$

Tensile test (uniaxial stress):

$$\sigma_1 > 0; \quad \sigma_2 = \sigma_3 = 0$$

$$\varepsilon_1 = \frac{\sigma_1}{E}; \quad \varepsilon_2 = \varepsilon_3 = -\nu\frac{\sigma_1}{E}$$

$$\nu = -\frac{\varepsilon_2}{\varepsilon_1} = -\frac{\varepsilon_3}{\varepsilon_1}$$

Yield Criteria

For plastic flow to occur, the combination of stresses must satisfy the yield criterion. Yield criteria have been formulated to describe the beginning of plastic deformation by relating the principal stresses to the tensile or compressive yield strength of the material.

- Tresca's yield criterion (maximum shear stress):

$$\tau_{\max} = \left(\sigma_{\max} - \sigma_{\min}\right)/2 = Y/2$$

$$\sigma_{\max} - \sigma_{\min} = Y$$

- von Mises' yield criterion (maximum strain energy):

$$\left(\sigma_1 - \sigma_2\right)^2 + \left(\sigma_2 - \sigma_3\right)^2 + \left(\sigma_3 - \sigma_1\right)^2 = 2Y^2$$

Plastic Region

Let's consider a prismatic workpiece whose initial dimensions are:

$$a_0 \quad b_0 \quad c_0$$

After a plastic deformation the final dimensions are:

$$a_1 \quad b_1 \quad c_1$$

Due to volume conservation, it follows that:

$$V = \text{constant} = a_0 \cdot b_0 \cdot c_0 = a_1 \cdot b_1 \cdot c_1 \qquad \frac{a_1 \cdot b_1 \cdot c_1}{a_0 \cdot b_0 \cdot c_0} = 1$$

$$\ln\left(\frac{a_1 \cdot b_1 \cdot c_1}{a_0 \cdot b_0 \cdot c_0}\right) = \ln 1 = 0$$

$$\ln\left(\frac{a_1}{a_0}\frac{b_1}{b_0}\frac{c_1}{c_0}\right) = \ln\left(\frac{a_1}{a_0}\right) + \ln\left(\frac{b_1}{b_0}\right) + \ln\left(\frac{c_1}{c_0}\right) = 0 \Rightarrow \varepsilon_1 + \varepsilon_2 + \varepsilon_3 = 0$$

Deformation of Solid Materials

Stress-strain relationship in plastic region

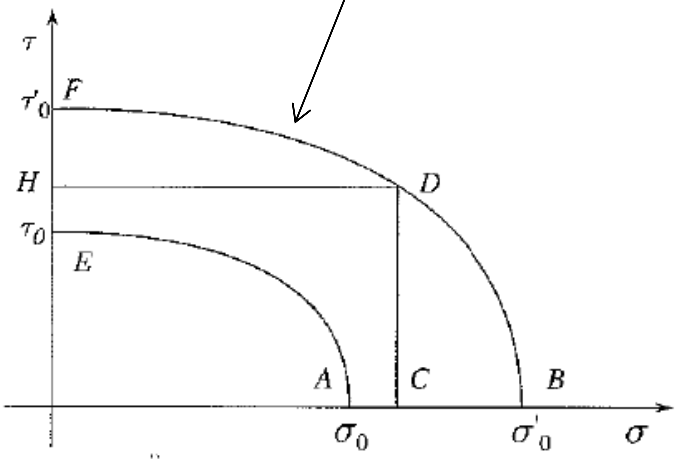
Mendelsson's experiment:

Very thin tubular specimen subject to tensile load and torsion load

Stresses:
- Normal stress, due to tensile load, σ_x.
- Shear stress, due to torsion load, τ_{xy}.

Two different loading paths:
- path 1: O-A-B-C-D
- path 2: O-E-F-H-D

von Mises' Yield Criterion



The same stress state D is reached, but with different permanent deformations:
- path 1: tensile strain
- path 2: shear strain

DEFORMATION OF SOLID MATERIALS

Stress-strain relationship in plastic region

Elastic Region

$$\varepsilon_1 = \frac{1}{E}\left[\sigma_1 - \nu\left(\sigma_2 + \sigma_3\right)\right]$$

$$\varepsilon_2 = \frac{1}{E}\left[\sigma_2 - \nu\left(\sigma_3 + \sigma_1\right)\right]$$

$$\varepsilon_3 = \frac{1}{E}\left[\sigma_3 - \nu\left(\sigma_1 + \sigma_2\right)\right]$$

$\Rightarrow$

Plastic region

$$d\varepsilon_1 = d\lambda\left[\sigma_1 - \frac{\sigma_2 + \sigma_3}{2}\right]$$

$$d\varepsilon_2 = d\lambda\left[\sigma_2 - \frac{\sigma_1 + \sigma_3}{2}\right]$$

$$d\varepsilon_3 = d\lambda\left[\sigma_3 - \frac{\sigma_1 + \sigma_2}{2}\right]$$

$$\varepsilon_1 + \varepsilon_2 + \varepsilon_3 = 0 \rightarrow \nu = \frac{1}{2}$$

$d\lambda$ considers the history of the deformation

Levy and von Mises proposed an expression for $d\lambda$

Stress-strain relationship in plastic region

Levy – von Mises equations

$$d\varepsilon_1 = \left(\sigma_1 - \frac{\sigma_2 + \sigma_3}{2}\right)\frac{d\bar{\varepsilon}}{\bar{\sigma}}$$

$$d\varepsilon_2 = \left(\sigma_2 - \frac{\sigma_1 + \sigma_3}{2}\right)\frac{d\bar{\varepsilon}}{\bar{\sigma}} \quad \Rightarrow \quad \frac{\bar{\sigma}}{d\bar{\varepsilon}} = \text{modulus of plasticity}$$

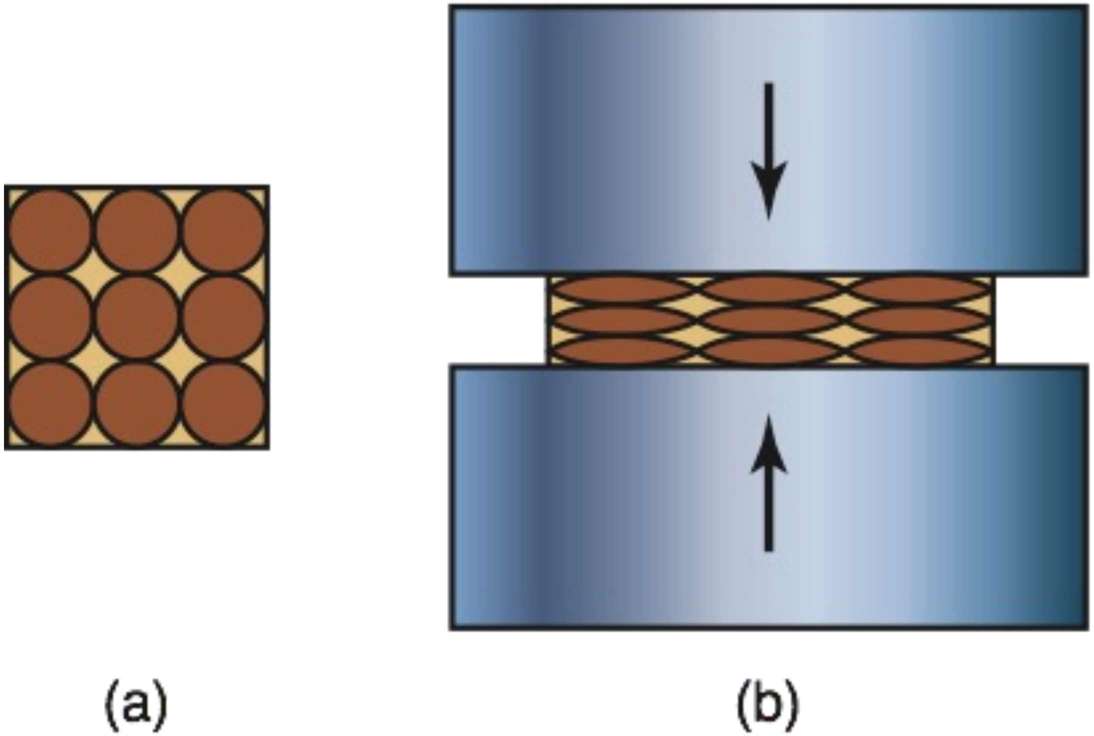
$$d\varepsilon_3 = \left(\sigma_3 - \frac{\sigma_1 + \sigma_2}{2}\right)\frac{d\bar{\varepsilon}}{\bar{\sigma}}$$

Where:

$$\bar{\sigma} = \frac{1}{\sqrt{2}}\left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2\right]^{1/2}$$ effective or equivalent stress

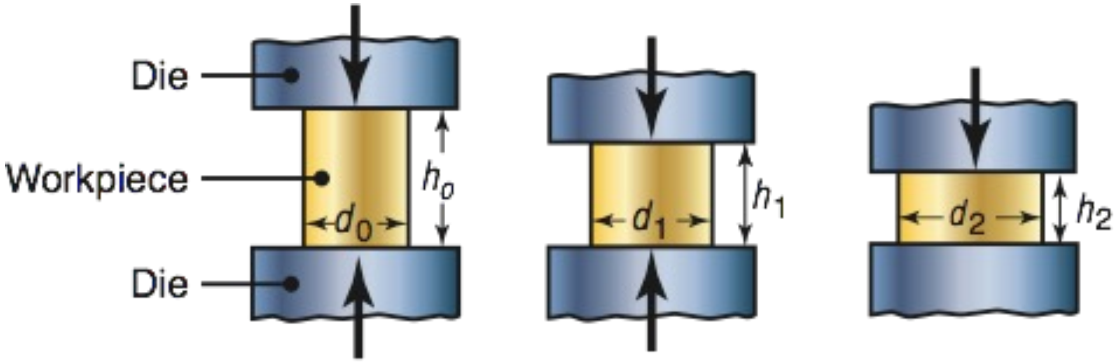
$$d\bar{\varepsilon} = \frac{\sqrt{2}}{3}\left[(d\varepsilon_1 - d\varepsilon_2)^2 + (d\varepsilon_2 - d\varepsilon_3)^2 + (d\varepsilon_3 - d\varepsilon_1)^2\right]^{1/2}$$ effective or equivalent strain

Compression load



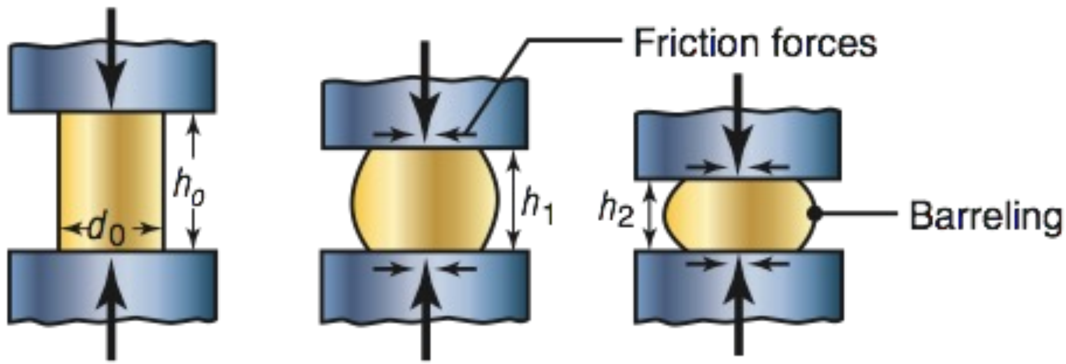
OPEN-DIE FORGING

Ideal compression with no friction:



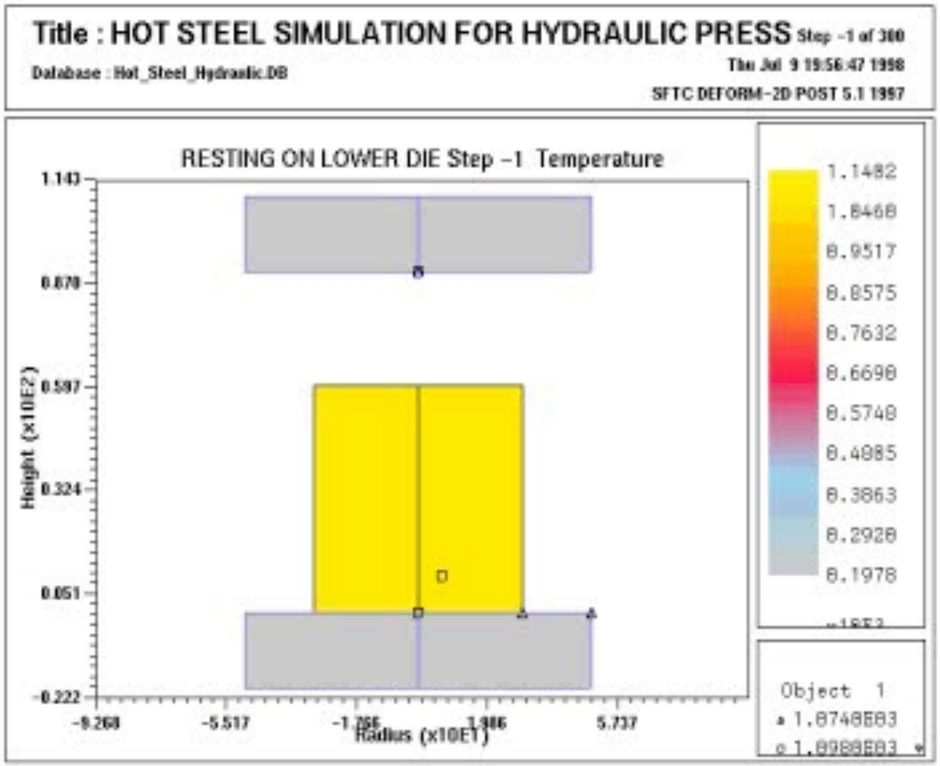
(a)

Real compression with friction:

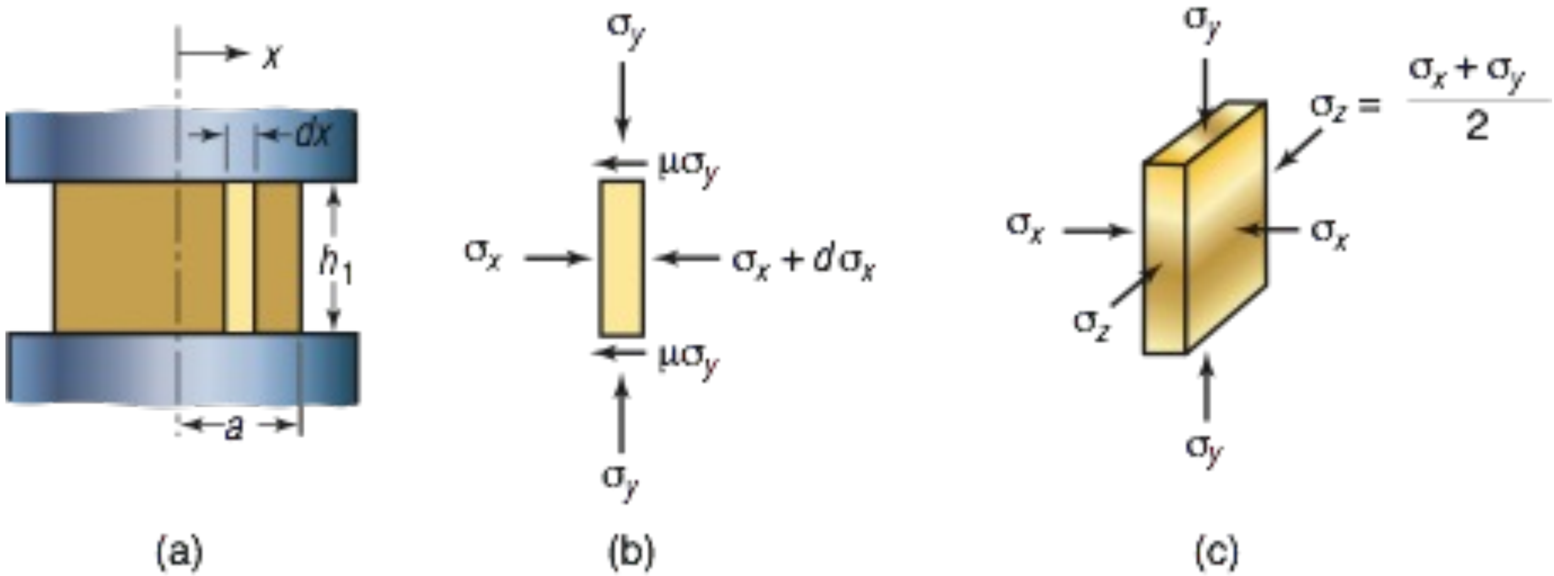


(b)

OPEN-DIE FORGING



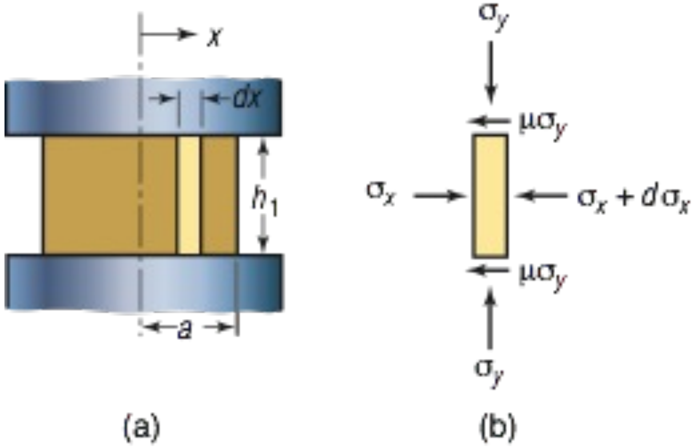
Slab method: plane strain condition



OPEN-DIE FORGING

Hypotheses:

- Material: perfectly plastic (no strain hardening)
- Plane strain (w = constant → $\varepsilon_z = 0$)
- Coefficient of friction
  μ = constant



Equilibrium conditions:

$$\sigma_y = p(x) = p$$

$$(\sigma_x + \partial\sigma_x)\cdot h\cdot w - \sigma_x\cdot h\cdot w + 2\mu\cdot p\cdot\partial x\cdot w = 0$$

$$\partial\sigma_x\cdot h + 2\mu\cdot p\cdot\partial x = 0$$

OPEN-DIE FORGING

Considering the plane strain condition and applying Levy – von Mises equations, it follows that:

$$\varepsilon_z = 0$$

$$\partial\varepsilon_z = \left(\sigma_z - \frac{\sigma_x + \sigma_y}{2}\right)\frac{\partial\bar{\varepsilon}}{\bar{\sigma}} = 0$$

$$\sigma_z - \frac{\sigma_x + \sigma_y}{2} = 0$$

$$\sigma_z = \frac{\sigma_x + \sigma_y}{2} = \frac{\sigma_x + p}{2}$$

Open-Die Forging

For plastic flow to occur, the combination of stresses must satisfy the yield criterion. Considering von Mises' yield criterion, it follows that:

$$(\sigma_x - \sigma_y)^2 + (\sigma_y - \sigma_z)^2 + (\sigma_z - \sigma_x)^2 = 2Y^2$$

$$(\sigma_x - p)^2 + \left(p - \frac{\sigma_x + p}{2}\right)^2 + \left(\frac{\sigma_x + p}{2} - \sigma_x\right)^2 = 2Y^2$$

$$\left[-(p - \sigma_x)\right]^2 + \left[\frac{1}{2}(p - \sigma_x)\right]^2 + \left[\frac{1}{2}(p - \sigma_x)\right]^2 = 2Y^2$$

$$(p - \sigma_x)^2 + \frac{1}{4}(p - \sigma_x)^2 + \frac{1}{4}(p - \sigma_x)^2 = 2Y^2$$

$$\frac{3}{2}(p - \sigma_x)^2 = 2Y^2 \rightarrow p - \sigma_x = \frac{2}{\sqrt{3}}Y$$

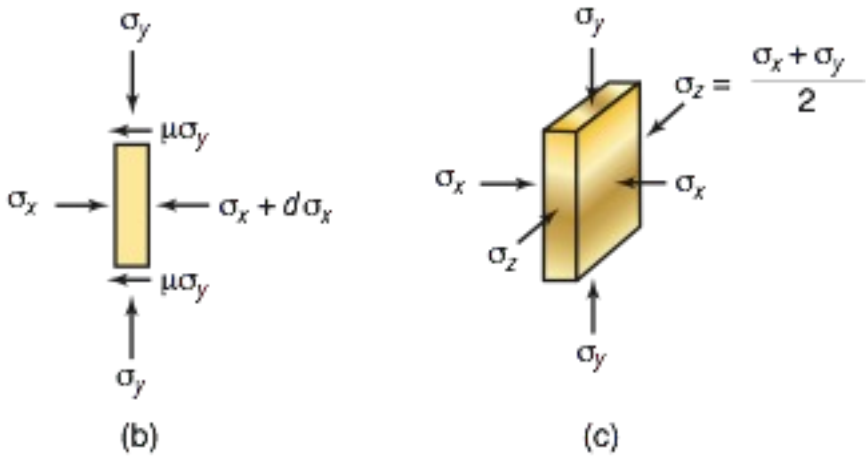
OPEN-DIE FORGING

Therefore:

$$p - \sigma_x = \frac{2}{\sqrt{3}} Y$$

$$\sigma_x = p - \frac{2}{\sqrt{3}} Y$$

$$\partial \sigma_x = \partial p$$



Then, the equilibrium condition becomes:

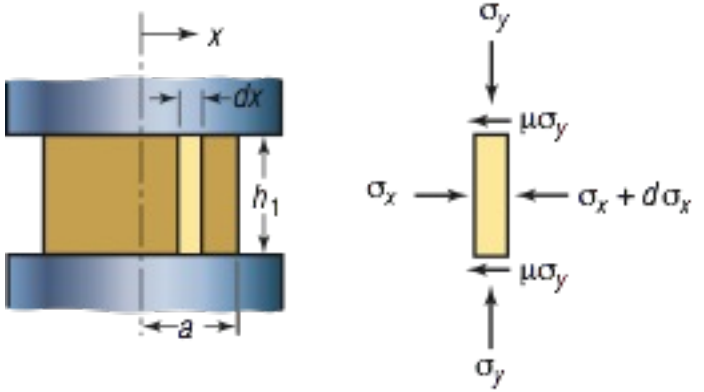
$$\partial \sigma_x \cdot h + 2\mu \cdot p \cdot \partial x = 0$$

$$\partial p \cdot h + 2\mu \cdot p \cdot \partial x = 0$$

$$\frac{2\mu}{h} \cdot \partial x = -\frac{\partial p}{p}$$

OPEN-DIE FORGING

Therefore:

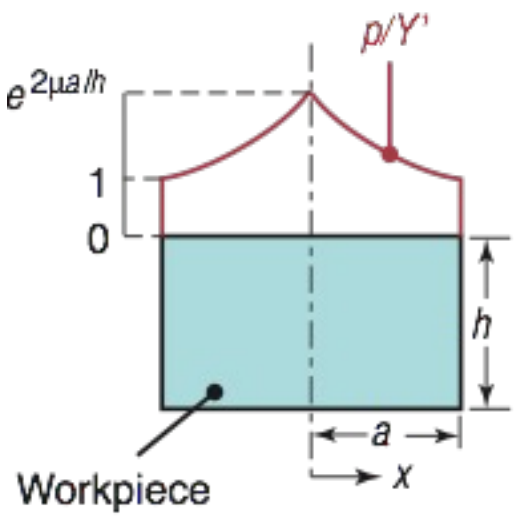


$$2\frac{\mu}{h}\cdot\int_x^a \partial x = -\int_p^{p_{external}} \frac{\partial p}{p}$$

$$2\frac{\mu}{h}\cdot\left(a-x\right) = \ln\frac{p}{p_{external}}$$

$$p = p_{external}\cdot e^{\frac{2\mu}{h}\left(a-x\right)} = \frac{2}{\sqrt{3}}Y\cdot e^{\frac{2\mu}{h}\left(a-x\right)}$$

$$\left[\sigma_x\left(x=a\right)=0 \rightarrow \sigma_y = p_{external} = \frac{2}{\sqrt{3}}Y = Y'\right]$$

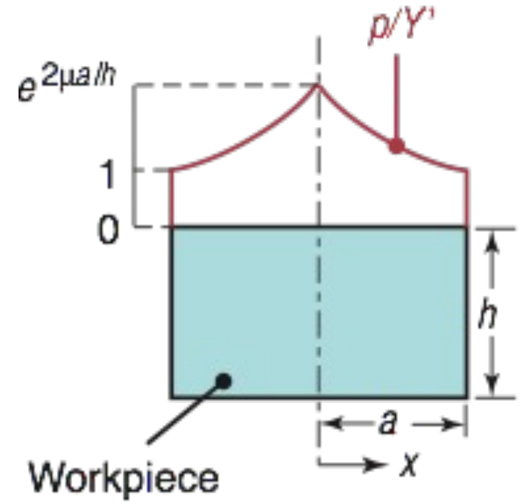


Friction hill
(die pressure variation)

Therefore, the average pressure is:

$$p_{av} = \frac{1}{a} \cdot \int_0^a p(x) \cdot \partial x = \frac{1}{a} \cdot \int_0^a \frac{2}{\sqrt{3}} Y \cdot e^{\frac{2\mu}{h}(a-x)} \cdot \partial x$$

$$p_{av} = \frac{2}{\sqrt{3}} Y \cdot \left(\frac{e^{\frac{\mu \cdot 2a}{h}} - 1}{\frac{\mu \cdot 2a}{h}} \right) \cong \frac{2}{\sqrt{3}} Y \cdot \left(1 + \frac{\mu a}{h} \right)$$

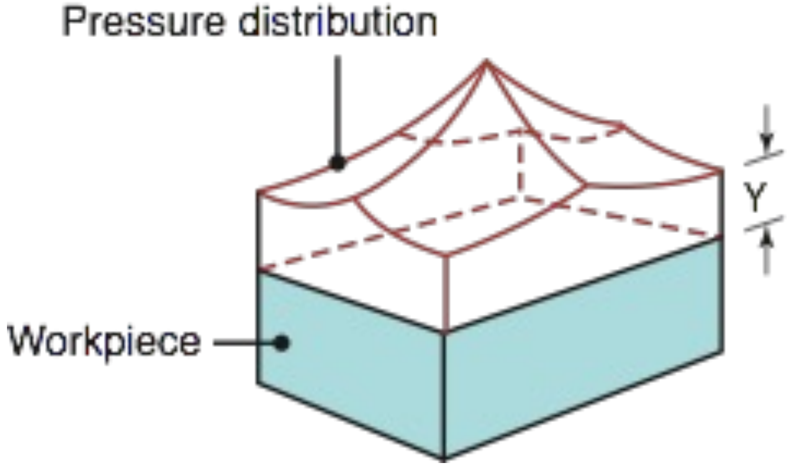


The force required to continue the compression at any given height h during the process can be obtained by:

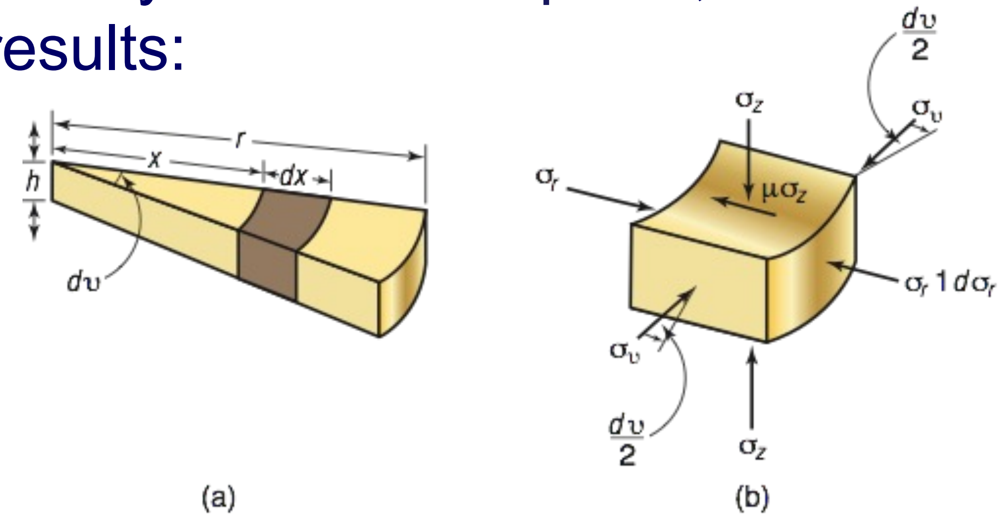
$$F = p_{av} \cdot 2a \cdot w \cong \frac{2}{\sqrt{3}} Y \cdot \left(1 + \frac{\mu a}{h} \right) \cdot 2a \cdot w$$

OPEN-DIE FORGING

Removing the hypothesis of plane strain, the result becomes:



Considering the deformation of a cylindrical workpiece, it results:



$$p(x) = Y \cdot e^{\frac{2\mu}{h}(r-x)}$$

$$p_{av} \cong Y\left(1 + \frac{\mu \cdot 2r}{3h}\right)$$

$$F = p_{av} \cdot \pi r^2$$

OPEN-DIE FORGING

Ideally, with no friction, the required work or energy (*W*) is:

- Prismatic workpiece (plane strain)

$$W = \int_{h_0}^{h_1} F(h)\cdot dh \cong \int_{h_0}^{h_1} \frac{2}{\sqrt{3}} Y \cdot 2a(h)\cdot w(h)\cdot dh$$

- Cylindrical workpiece

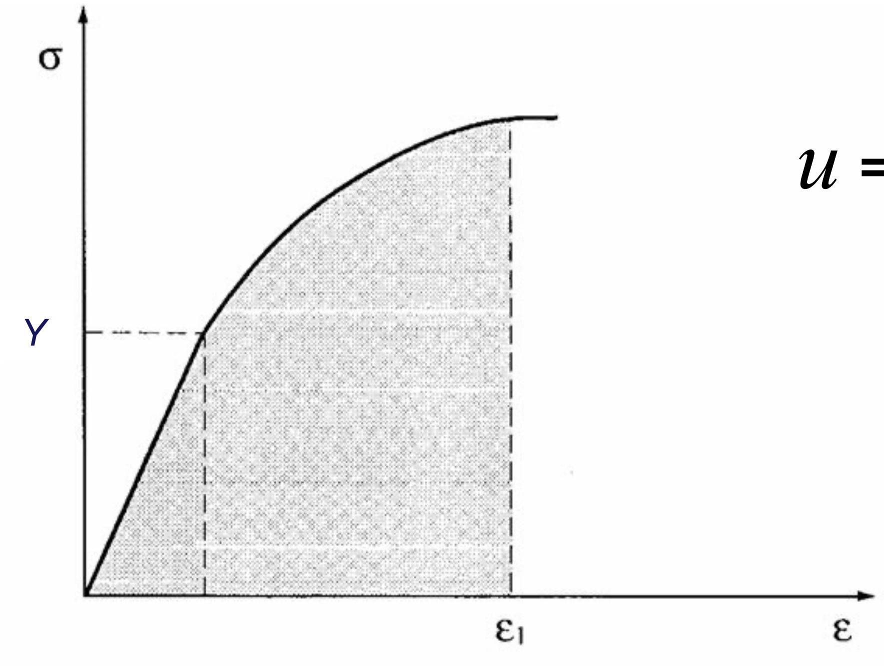
$$W = \int_{h_0}^{h_1} F(h)\cdot dh \cong \int_{h_0}^{h_1} Y \cdot \pi r^2(h)\, dh$$

$$W = \int_{h_0}^{h_1} Y \cdot \underbrace{\pi r^2 h}_{V} \frac{dh}{h} = VY \ln\left[h\right]_{h_0}^{h_1} = VY\varepsilon = Vu$$

STRAIN ENERGY

The energy stored in a body due to deformation is called the strain energy. The strain energy per unit volume is called the **strain energy density** (u).

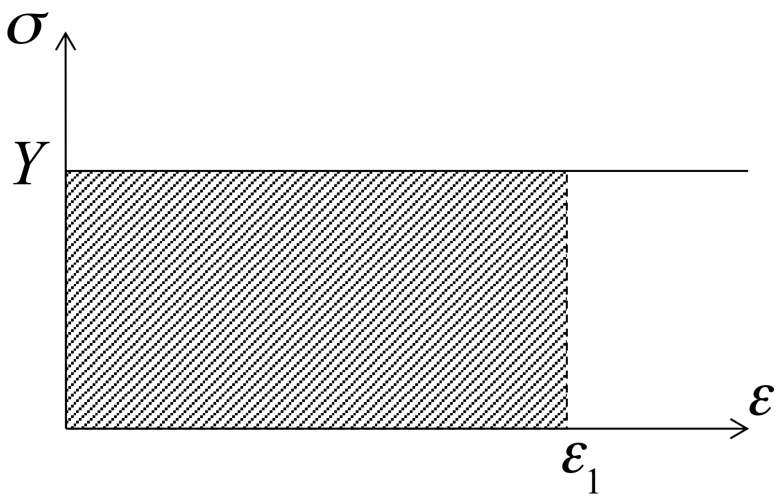
Considering the tensile test (uniaxial stress) it is the area underneath the stress-strain curve up to the point of deformation (ε_1).



$$u = \int_0^{\varepsilon_1} \sigma \cdot d\varepsilon$$

STRAIN ENERGY

Considering a **perfectly plastic** material the strain energy density (u) results:



$$u = \int_0^{\varepsilon_1} \sigma \cdot d\varepsilon$$

$$u = \int_0^{\varepsilon_1} Y \cdot d\varepsilon$$

$$u = Y \cdot \varepsilon_1$$

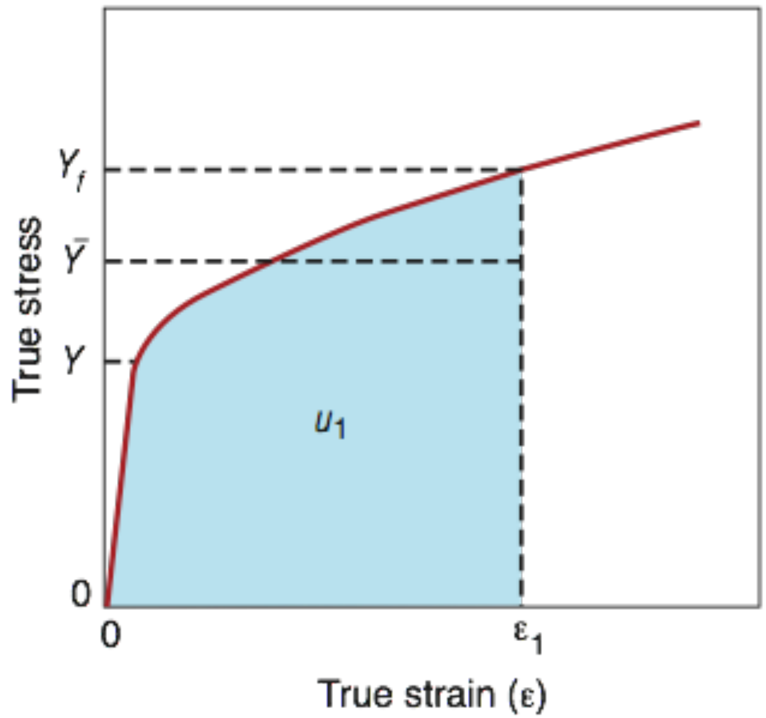
STRAIN ENERGY

Considering a true stress – true strain relationship with **strain hardening**, the strain energy density (u) results:

$$u = \int_0^{\varepsilon_1} \sigma \cdot d\varepsilon$$

$$\sigma = k \cdot \varepsilon^n$$

$$u = \int_0^{\varepsilon_1} k \cdot \varepsilon^n \cdot d\varepsilon = \frac{k \cdot \varepsilon_1^{n+1}}{n+1} = \bar{Y}_f \cdot \varepsilon_1$$



Where: $\bar{Y}_f = \dfrac{k \cdot \varepsilon_1^n}{n+1}$ is the average flow stress or mean flow stress.

Work of Deformation

Starting from the strain energy density it is possible to calculate the strain energy (*U*) that is equal to the **work of deformation** (*W*) done by external forces to form the material:

$$U = W = u \cdot V$$

where *V* is the volume of the workpiece.

This is the ideal work of deformation. In fact it does consider the strain energy density, but does not consider friction and redundant deformation.

The total work of deformation can be calculated as:

$$W_{tot} = u_{tot} \cdot V$$

$$u_{tot} = u + u_f + u_r$$

Where:

u_f is the frictional work per unit of volume

u_r is the redundant work per unit of volume

WORK OF DEFORMATION

Considering a triaxial stress state, the strain energy density may be determined as follows:

$$du = \sigma_1 \cdot d\varepsilon_1 + \sigma_2 \cdot d\varepsilon_2 + \sigma_3 \cdot d\varepsilon_3$$

It is also possible to determine it starting from the effective or equivalent stress and strain as follows:

$$u = \int_0^{\varepsilon} \overline{\sigma} \cdot d\overline{\varepsilon}$$

SUMMARY

- Bulk deformation processes: open-die forging, impression-die forging, and flashless forging
- Sheet metalworking: bending, deep drawing, and shearing processes
- 3D stress-strain relationship
  - Principal axes and principal stresses
  - Elastic region: Hooke's law for triaxial stress
  - Yield Criteria: Tresca and von Mises
  - Plastic region: Levy – von Mises equations
- Slab method for open-die forging